

Ph.D. ENTRANCE EXAMINATION, APRIL 2022

COMPUTER SCIENCE

Time : Two Hours

Maximum : 100 Marks

Part A

*Answer all questions.**Each question carries 2 marks.*1. What is the infix expression for $+ \cdot^* abcd // ef + gh$?

A) $((((a \wedge b) * c) - d) / (e / f) + (g + h).$

B) $((((a \wedge b) * c) - d) + (e / f) + (g + /).$

C) $((((a * b) \wedge c) d) + (e / f) / (g + h).$

D) $((((a \wedge b) * c) - d) + (e / f) / (g + h).$

2. How many passes of bubble sort are required to sort the following sequence (Pass is counted only when at least one swap is performed in the bubble sort pass) ?

`int Array[] = {25, 21, 6, 15, 18, 20, 14, 17, 16}`

A) 8.

B) 7.

C) 9.

D) 5.

3. The postorder traversal of a binary tree is 8, 9, 6, 7, 4, 5, 2, 3, 1. The inorder traversal of the same tree is 8, 6, 9, 4, 7, 2, 5, 1, 3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

A) 2.

B) 3.

C) 4.

D) 5.

4. A Selective Repeat ARQ and Go-Back-n ARQ uses a window of size 31. The number of bits needed to define sequence numbers respectively are :

A) 6, 5.

B) 6, 6.

C) 5, 6.

D) 5, 5.

Turn over

5. The Mobile Application Protocol (MAP) typically runs on top of which protocol :

- A) SNMP (Simple Network Management Protocol).
- B) SMTP (Simple Mail Transfer Protocol).
- C) SS7 (Signalling System 7).
- D) HTTP (Hyper Text Transfer Protocol).

6. Advanced Encryption Standard (AES) is based on :

- A) Asymmetric key algorithm.
- B) Symmetric key algorithm.
- C) Public key algorithm.
- D) Key exchange.

7. Consider the following regular expression :

$$R = (aa)^* (bb^*)^* b$$

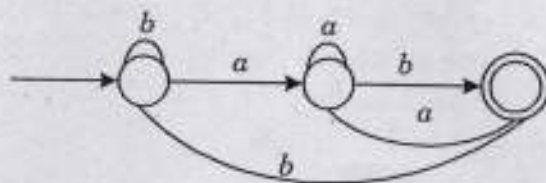
Select the correct language generated by R from the given options :

- A) $L = \{a^{2m}b^{2n+1} \mid m, n \geq 0\}$.
- B) $L = \{a^{2m}b^n \mid m, n \geq 0\}$.
- C) $L = \{a^{2m}b^{n+1} \mid m, n \geq 0\}$.
- D) None of these.

8. Context-free languages are :

- A) Closed under union, intersection.
- B) Closed under complementation, Kleene closure.
- C) Closed under intersection, complement.
- D) Closed under union, Kleene closure.

9. If the final states and non-final states in the DFA below are interchanged, then which of the following languages over the alphabet $\{a, b\}$ will be accepted by the new DFA :



- A) Set of all strings that do not end with ab .
- B) Set of all strings that begin with either an a or a b .
- C) Set of all strings that do not contain the substring ab .
- D) The set described by the regular expression $b^*aa^*(ba)^*b^*$.

- $$\begin{array}{l} A \rightarrow B \\ B \rightarrow D \\ C \rightarrow DE \\ CD \rightarrow AB \end{array}$$

A) C₊
B) (CD)₊
C) A₊
D) (CA)₊

- A) 51.
C) 53.
- B) 52.
D) 54.

- A) Key Constraint.
B) Foreign Key Constraint.
C) Entity Integrity Constraint.
D) Domain Constraint.

- A) 5473364. B) 5123136.
C) 5473364. D) 4573264.

- A) Global variables but not heap.
- B) Heap but not global variables.
- C) Neither global variables nor heap.
- D) Both heap and global variables.

15. In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed ?
- I. Contiguous.
 - II. Linked.
 - III. Indexed.
- A) I and III only. B) II only.
C) III only. D) II and III only.
16. Let p , q and r be the propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is :
- A) A tautology.
 - B) A contradiction.
 - C) Always TRUE when p is FALSE.
 - D) Always TRUE when q is TRUE.
17. Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _____.
- A) 24. B) 20.
C) 32. D) 64.
18. The term "TRP" is associated with TV shows stands for :
- A) Total Rating Points.
 - B) Time Rating Points.
 - C) Thematic Rating Points.
 - D) Television Rating Points.
19. Shadow mask technology is used in the manufacturing of which device :
- A) LCD.
 - B) CRT.
 - C) Light Pen.
 - D) Plasma Display.

20. Research ethics do not include :

- A) Honesty.
- B) Subjectivity.
- C) Integrity.
- D) Objectivity.

21. 'Sampling Cases' means :

- A) Sampling using a sampling frame.
- B) Identifying people who are suitable for research.
- C) Literally the researcher's brief case.
- D) Sampling of people, newspapers, television programmes etc.

22. What's going to happen when we compile and run the following C program ?

```
#include "stdio.h"

int main()
{
    int i = 1, j ;
    for (;;)
    {
        if(i)
            j = --i ;
        if (j < 10)
            printf("welcome", j++);
        else
            break ;
    }
    return 0 ;
}
```

- A) Compile error.
- B) No compile error but it will run into infinite loop printing welcome.
- C) No compile error and it'll print welcome 10 time.
- D) No compile error but it'll print welcome 9 times.

23. Which of the following statement is correct for **switch** controlling expression ?
- A) Only int can be used in "switch" control expression.
 - B) Both int and char can be used in "switch" control expression.
 - C) All types i.e. int, char and float can be used in "switch" control expression.
 - D) "switch" control expression can be empty as well.
24. Research problem is selected from the stand point of :
- A) Researcher's interest.
 - B) Financial support.
 - C) Social relevance.
 - D) Availability of relevant literature.
25. A computer has six tape drives, with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free :
- A) 6.
 - B) 5.
 - C) 4.
 - D) 3.

(25 × 2 = 50 marks)

Part B

*Answer any five questions.
Each question carries 10 marks.*

1. a) State any five factors to be considered while preparing a research design. (5 marks)
b) Discuss the basic concepts of testing the hypothesis. (5 marks)
2. a) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q$,
 $Q \rightarrow R$, $P \rightarrow M$, and $\neg M$. (6 marks)
b) Give examples of sets A, B, C such that $A \cup B = A \cup C$, but $B \neq C$. (4 marks)
3. a) Explain about image sampling and quantization process. (5 marks)
b) Discuss the basic steps involved in Canny edge detection. (5 marks)

4. a) Consider the set of 6 processes whose arrival time and burst time are given below. If the CPU scheduling policy is Round Robin with time quantum = 3, calculate the average waiting time and average turn around time :

Process ID	Arrival Time	Burst Time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

(5 marks)

- b) Explain how semaphores can be used to deal with n -process critical section.

(5 marks)

5. a) Suppose variable names in a programming language can be either a single uppercase letter or an uppercase letter followed by a digit. Find the number of possible variable names.

(4 marks)

- b) Discuss any two applications of graph theory in Computer Science.

(6 marks)

6. a) Explain asymptotic notations.

(4 marks)

- b) Construct an AVL tree by inserting the following elements in the given order.

63, 9, 19, 27, 18, 108, 99, 81.

(6 marks)

7. a) Derive the time complexity of quick sort in average case.

(5 marks)

- b) List any three properties of Discrete Fourier Transform.

(5 marks)

8. a) Discuss about plagiarism in research.

(5 marks)

- b) In a group of 50 students 24 like cold drinks and 36 like hot drinks and each student likes at least one of the two drinks. How many like both coffee and tea ?

(5 marks)

[5 × 10 = 50 marks]